GABRIEL REYNA ALVARADO

Data Analyst | Machine Learning

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Profile

Computer Science student with hands-on experience across the full data project lifecycle: dataset preparation and cleaning, training and evaluation of classification models, and deployment of results in dashboards for non-technical users (Tableau and Streamlit). I work with explicit methodological rigour — data leakage detection, metric selection driven by the cost of each error type, and documented limitations — on a Python, SQL and scikit-learn/XGBoost stack. Every project below links to its code and to a write-up of the decisions behind it.

Data and Machine Learning Projects

ALDIMI Predict — Predictive System for a Paediatric Oncology Shelter

Python, XGBoost, Scikit-learn, SHAP, SQLite, Streamlit | UPC, 2026 | Team of four: modelling and deployment

- Trained and evaluated classification models (XGBoost, Random Forest) to anticipate 7- and 14-day stockouts in the inventory of a shelter for paediatric oncology patients, tuning hyperparameters with RandomizedSearchCV over 5-fold StratifiedKFold.
- Built a multiclass clinical risk classifier over 34 nutritional, socioeconomic and treatment-adherence variables, reaching a macro F1 of 0.75 and AUC of 0.87 on a synthetic dataset generated to protect the confidentiality of real patients.
- Chose macro F1 over accuracy because the risk classes are imbalanced and a missed high-risk case costs far more than a false positive.
- Interpreted predictions with SHAP values to surface the dominant risk factors, enabling shelter staff to audit a model that informs care decisions.
- Deployed the system as a Streamlit dashboard over a unified SQLite database, delivered with a user manual and a data dictionary.

Smart Kitchen Intelligence — Food Waste Analytics

Python, Polars, Pandas, Scikit-learn, Tableau | UPC, 2026 | Team of three: data preparation, modelling and visualisation

- Implemented the cleaning pipeline over 25,819 inventory movements, applying 6 verifiable quality rules (outliers, category harmonisation, referential integrity, structural nulls, deduplication), with cross-implementation in Polars and Pandas for independent validation.
- Trained waste classification models (Random Forest against Logistic Regression), detecting and removing a data leak that inflated metrics to F1 = 1.0, and selected the final model on F1 and cross-validation stability rather than accuracy, since a trivial classifier already reached 64.5%.
- Designed a 10-sheet Tableau dashboard with 4 KPIs, a global household filter and a prescriptive panel, which identified that 97% of the economic loss concentrated in a single storage location, with a waste rate 9.1 times higher than the most efficient one.
- Validated the representativeness of the simulated dataset against the UNEP Food Waste Index Report 2024 benchmark, and documented the structural redundancy between location and product category as a methodological limitation.

Anti-Waste Recommendation Engine — Hybrid Recommender

Python, Scikit-learn, ALS, NetworkX, Streamlit | UPC, 2026 | Team of three: modelling and deployment

- Trained a three-signal hybrid recommender combining TF-IDF content similarity, implicit ALS collaborative filtering, and expiry-proximity urgency, evaluated under a blind masked-basket completion protocol and achieving 100% catalogue coverage against the popularity baseline.
- Segmented consumption patterns with DBSCAN after benchmarking against K-Means and GMM on the PCA-reduced space, reaching a Silhouette Score of 0.65 with under 0.3% noise.
- Empirically ruled out adding PageRank centrality to the ensemble through an ablation over the weight grid (optimal weight of zero), documented the negative result in the technical report, and deployed the system as a Streamlit application.

FruitGuard — Produce Freshness Classification with Computer Vision

Python, OpenCV, Scikit-learn, Scikit-image, Gradio | UPC Winter School, 2025 | Team of three: modelling and evaluation

- Built a freshness classifier for nine produce types over 30,357 images, combining HOG, GLCM and colour features with an OpenCV preprocessing pipeline (normalisation and noise reduction), reaching 97.4% test accuracy across 14 evaluated classes.

Professional Experience

Data Analyst Intern — Cirion Next Generation

Lima, Peru | January 2024 – March 2024

- Processed and cleaned datasets of over 10,000 records using SQL and Python (Pandas), implementing consistency checks that eliminated manual rework in engineering reports.
- Automated the generation of recurring visualisations with Matplotlib and Seaborn, freeing roughly 20 hours of manual work per month for the team.
- Refactored SQL queries alongside the data engineering team, reducing execution times for business-critical reports.

Founder — Camote Studio

Lima, Peru | 2025 – Present

- Co-founded an independent game studio and shipped *Space Drunks*, a cartoon-styled beat 'em up, working across development, production and team coordination rather than a single fixed role.
- Established the Git version control workflow for a repository of over 500 binary assets, eliminating integration conflicts during team work.

Technical Skills

Languages: Python, SQL, Java, C++, R, JavaScript

Data and Machine Learning: Pandas, Polars, NumPy, Scikit-learn, XGBoost, SHAP, OpenCV

Techniques: Supervised classification, clustering (DBSCAN, K-Means, GMM), PCA, t-SNE, collaborative filtering (ALS), TF-IDF, cross-validation

Visualisation: Tableau, Streamlit, Matplotlib, Seaborn, Plotly

Databases: SQL Server, MySQL, SQLite, MongoDB, relational modelling

Tools: Git, GitHub, Docker, pytest, Excel

Methodologies: CRISP-DM, Scrum

Education

BSc in Computer Science — Universidad Peruana de Ciencias Aplicadas (UPC)

Lima, Peru | Expected graduation: 2026

Relevant coursework: Machine Learning, Data Visualisation, Database Management, Statistical Analysis.

Computer Vision — Artificial Intelligence

UPC Winter School, with the University of Pittsburgh | July 2025

Five-day intensive taught by Nils Murrugarra-Llerena, of Pittsburgh's School of Computing and Information. Where the FruitGuard classifier was built.

Game Development Workshop — UPC GameLab

Lima, Peru | 2024 – 2025

Term-long workshop where teams form and ship a game; the team behind Camote Studio came out of it. Modelled real-time simulation systems with object-oriented programming and optimised data structures, reducing processing overhead by 15% against the baseline implementation.

Languages, Certifications and Activities

Languages: Spanish (native), English (B2 — Cambridge First, 2025), French (intermediate)

Certifications: Scrum Fundamentals Certified (SFC), SCRUMstudy, 2023 | Introduction to MongoDB, 2023 | Pixel Art for Video Games, Michigan State University via Coursera, 2024
| LinkedIn Certifications

Activities: PMI Lima Peru Chapter; PMI LATAM Conference 2026. Inter-university innovation competition on SDG 11 (Sustainable Cities and Communities): member of a team of 6 that developed and defended a proposal before an industry jury.